
Markov Chain Lock Watermarking: A Calibrated Policy-to-Configuration Map for EU AI Act Article 50

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Abstract

Regulatory regimes such as the EU AI Act mandate machine-readable marking of synthetic text, but existing watermark detectors rely on the generating LM and on heuristic thresholds with no closed-form calibration. We introduce *Markov Chain Lock* (MCL), an active watermark that partitions the vocabulary into S states via keyed SHA-256 and forces a hard Markov transition on a fraction ρ of positions; the detector replays the partition from the same key in $O(n)$ hash operations, with no LM access. We derive a closed-form $S^*(n, \rho, \alpha)$ mapping a target FPR, text length, and budget to the minimum state count ([Theorem A.2](#)), prove a universal robustness threshold $\delta^* = 1 - 1/\sqrt{2} \approx 29.3\%$ that is invariant in (S, ρ, n) ([Theorem A.4](#)), and generalise both to any k -regular transition topology ([Theorem A.6](#)). Across three instruction-tuned LLMs and four domains, MCL strictly dominates KGW and SWEET under translation and random-substitution attacks at matched budget; an empirical-SD threshold recalibration restores the 1% target FPR on natural-language text ([section 6](#)).

1. Introduction

EU AI Act Article 50 ([European Parliament and Council, 2024](#)) requires a machine-readable mark on AI-generated text by August 2026, but does not specify which false-positive rate, text length, or quality cost constitute a compliant detector. Operators therefore need a watermark whose detector is auditable end-to-end. A regulator or third-party verifier should be able to confirm provenance from the text, the secret key, and the (publicly known) tokenizer alone — without query access to the generating language model.

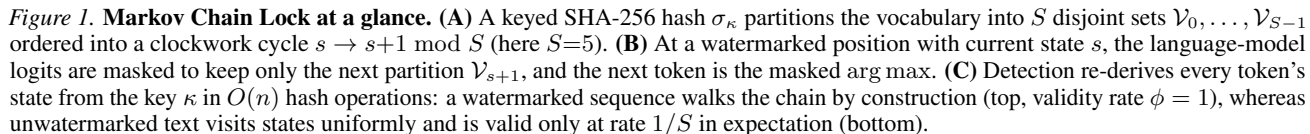
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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

The widely used logit-bias schemes do not meet this bar. KGW ([Kirchenbauer et al., 2023](#)) and its entropy-gated variants such as SWEET ([Lee et al., 2024](#)) and EWD ([Lu et al., 2024](#)) require the language model (or a faithful proxy) at detection time in order to recompute the green-list bias or to weight tokens by entropy. Undetectability-frontier results ([Christ et al., 2024](#)) characterise what any scheme can achieve information-theoretically but yield no deployable construction, and passive detectors ([Mitchell et al., 2023](#); [Tian, 2023](#)) expose no controllable false-positive frontier at all.

Markov Chain Lock. We propose *Markov Chain Lock* (MCL), an active watermark whose detector is model-free. A keyed SHA-256 hash partitions the vocabulary into S equivalence classes (states), and a fixed cycle over those states defines a Markov transition pattern. On a fraction ρ of generated positions, the language-model logits are masked to keep only tokens whose state is the legal successor of the current state; the produced text therefore walks the chain by construction. Detection re-derives every token’s state from the key in $O(n)$ hash operations and counts how often consecutive tokens follow the cycle. Because the state map is a function of the token id and the key only, no language-model forward pass, weights, or proxy LM is required to verify a transcript. Two operational consequences follow. First, third-party audit becomes lightweight: a regulator with the key can verify provenance from a string of plain text. Second, the detector’s null-and-alternative law inverts in closed form to a calibration map ([Theorem A.2](#), derived in [section A](#)). A regulator who fixes a target false-positive rate α , a text-length floor $n_{\min}$, and a watermark budget ρ reads off the minimum admissible state count $S^*(n_{\min}, \rho, \alpha)$; a deployer may spend more budget than the floor for additional margin but cannot evade the regulator’s specification. [Figure 1](#) summarises the construction; the formal treatment is in [section 4](#) and [section 5](#).

Gating as a design dial. A second design choice — *which* of the n positions to watermark — is left as a parameter we call the gate. Existing schemes implicitly fix this choice: SWEET and EWD watermark only high-entropy positions, on the intuition that those positions contribute the



Contributions.

- Roadmap.** [section 2](#) situates MCL within active and passive watermarking; [section 3](#) fixes notation; [section 4](#) specifies the construction; [section 5](#) states the calibration, robustness, and k -regular results; and [section 6](#) reports the empirical comparison. Proofs and additional derivations are in [section A](#).

Passive detection. DetectGPT (Mitchell et al., 2023) exploits log-probability curvature; GPTZero (Tian, 2023) combines perplexity and burstiness. Both read statistical traces in existing text, with no controllable false-positive frontier and sharp degradation under paraphrase (Krishna et al., 2023; Sadasivan et al., 2023). They cannot serve as the technical substrate for Article 50 because neither regulator nor deployer can set the detection regime in advance.

Distortion-free schemes. Aaronson (2023)’s Gumbel-max construction and Kuditipudi et al. (2024)’s inverse-transform sampling watermark are *distortion-free*: they preserve the LM marginal in expectation over the key. Christ et al. (2024) prove cryptographic indistinguishability under sufficient entropy. These schemes beat MCL on quality strictly (zero expected KL versus MCL’s $\geq \rho \log S$). The trade is that they require either LM access or the key se-

quence at detection time; MCL is the distorting alternative that buys model-free third-party audit and the closed-form $(\alpha, n, \rho) \mapsto S^*$ map.

Entropy-adaptive schemes: gates vs. detectors. SWEET (Lee et al., 2024) and EWD (Lu et al., 2024) both restrict watermarking signal to *high-entropy* positions, but at different layers. SWEET applies a binary gate at generation time (restricting where the green-list bias is added); EWD weights positions by entropy at detection time, approximating the log-likelihood ratio of Theorem A.9. MCL is gate-agnostic in its mathematical specification: high-entropy, low-entropy, and surprisal-gap gates are all admissible. For the empirical comparison in section 6 we adopt the high-entropy gate, matching SWEET’s gating policy at the same budget; the operational dial we vary in our matched-budget head-to-head is therefore ρ , not the gate identity.

Technical AI governance. Article 50 of the EU AI Act (European Parliament and Council, 2024) and the EU’s draft Code of Practice on Transparency (European Commission AI Office, 2025) require machine-readable marking of AI-generated content; the OECD Hiroshima Process (OECD, 2024) provides a parallel international framework. These instruments leave the detection regime (FPR, quality floor, robustness) implicit. MCL’s contribution, beyond the watermarking scheme itself, is that it surfaces these parameters analytically, producing a policy-to-configuration map auditable by third parties (section 7).

3. Preliminaries

Notation. Let $\mathcal{V}$ denote the vocabulary of an autoregressive language model, $V = |\mathcal{V}|$, and write $\mathbf{t} = (t_1, \dots, t_n) \in \mathcal{V}^n$ for a token sequence of length n . The deployer and the verifier share a secret key $\kappa \in \{0, 1\}^*$ and the same tokenizer; the verifier never sees the language model. Let $\mathcal{H} : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$ be SHA-256, modelled as a random oracle in the security analysis (Theorem A.8). We write $[m] = \{0, \dots, m-1\}$ and z_α for the one-sided standard-normal quantile at level α .

Threat model. The adversary observes watermarked text and may paraphrase, translate, edit, splice, or delete tokens before publication; we model these as a per-token modification rate $\delta \in [0, 1]$. The adversary is computationally bounded and has no access to κ or to the language model at detection time. The verifier sees only token text and runs in $O(n)$ hash operations.

Detection problem and quality. Detection is a binary hypothesis test on text alone— H_0 (non-watermarked) vs. H_1 (watermarked under key κ)—at a fixed false-positive rate α , since false positives are operationally the costly error.

The watermark distribution must remain fluent under standard greedy or low-temperature decoding, which we control via the per-token KL divergence to the original next-token distribution and report as perplexity inflation (section 4).

4. The MCL Scheme

4.1. State partition

For each token id $v \in \mathcal{V}$ define

$$\sigma_\kappa(v) = \mathcal{H}(\kappa \| v) \bmod S, \quad (1)$$

which deterministically maps $\mathcal{V}$ onto S equivalence classes $\mathcal{V}_s = \{v : \sigma_\kappa(v) = s\}$, $s \in [S]$. Under the random-oracle model σ_κ is uniform and independent across distinct tokens, so $\mathbb{E}[|\mathcal{V}_s|] = V/S$; knowledge of σ_κ on queried tokens reveals nothing about unqueried ones (Theorem A.8).

4.2. Transition topology

A successor function $\Sigma_k : [S] \rightarrow \binom{[S]}{k}$ assigns each state a k -element set of legal next states. We require Σ_k to be k -regular in both out- and in-degree; column-regularity is essential for the null-variance and robustness arguments (Theorem A.6, Theorem A.1). The default is the *clockwork* chain at $k = 1$, $\Sigma_1(s) = \{(s+1) \bmod S\}$: deterministic, periodic of period S , uniform stationary distribution, and the smallest random baseline $1/S$ among 1-regular topologies. We additionally evaluate the *soft-cycle* variant $k = 2$, $\Sigma_2(s) = \{(s+1) \bmod S, (s+2) \bmod S\}$, which doubles the per-state successor budget at the cost of raising the random baseline to $2/S$ (section 6).

4.3. Generation

At step i with current state $s_i = \sigma_\kappa(t_i)$, MCL forms the legal token set $\mathcal{V}_{s_i}^* = \bigcup_{s' \in \Sigma_k(s_i)} \mathcal{V}_{s'}$ and masks the LM logits outside $\mathcal{V}_{s_i}^*$ to $-\infty$. At gated positions MCL picks the argmax over the masked logits; at ungated positions it samples from the original distribution. Each step costs one LM forward pass plus an $O(V)$ logit mask (precomputed once per state).

4.4. Watermark budget and entropy gate

The mask is applied only at a subset of positions; the *watermark budget* is $\rho = \mathbb{E}_i[g_i] \in [0, 1]$. Let $H_i = -\sum_t p_i(t) \log p_i(t)$ be the next-token entropy and $\Delta_i = p_i^{(1)} - p_i^{(2)}$ the gap between the top two probabilities.

Definition 4.1 (Gate family). $G_{\text{all}}: g_i = 1$; $G_{H_{\text{high}}}(\tau): g_i = \mathbb{1}[H_i > \tau]$; $G_{H_{\text{low}}}(\tau): g_i = \mathbb{1}[H_i < \tau]$; $G_\Delta(\tau): g_i = \mathbb{1}[\Delta_i < \tau]$.

Our default is $G_{H_{\text{high}}}$, which masks only the model’s uncertain positions and matches the high-entropy gating policy

Algorithm 1 MCL Watermark Embedding

Input: prompt p , key κ , states S , successor Σ_k , gate $g(\cdot)$, max tokens N

$\mathbf{t} \leftarrow \text{Tokenize}(p); s \leftarrow \sigma_\kappa(\mathbf{t}_{-1})$

for $i = 1$ **to** N **do**

$p_i \leftarrow \text{softmax}(\mathcal{M}(\mathbf{t}))$

$\mathcal{V}^* \leftarrow \bigcup_{s' \in \Sigma_k(s)} \mathcal{V}_{s'}$

if $g(p_i) = 1$ **and** $p_i(\mathcal{V}^*) > 0$ **then**

$t_i \leftarrow \arg \max_{t \in \mathcal{V}^*} p_i(t)$

else

$t_i \leftarrow \arg \max_t p_i(t)$

end if

$\mathbf{t} \leftarrow \mathbf{t} \parallel t_i; s \leftarrow \sigma_\kappa(t_i)$

if $t_i = \text{EOS}$ **then**

break

end if

end for

return $\text{Decode}(\mathbf{t})$

of SWEET (Lee et al., 2024) at the same budget; section 6 reports the head-to-head. The threshold τ is calibrated by quantile-matching on a held-out pilot so the realised gate rate tracks ρ . $G_{H_{\text{low}}}$ is the *anti-SWEET* ablation, and G_Δ targets near-tied top-two positions where the greedy substitution regret is bounded pointwise by Δ_i .

4.5. Detection

The verifier tokenises the candidate text, re-derives states by re-applying σ_κ , and computes the fraction of legal transitions

$$\phi(\mathbf{t}) = \frac{1}{n-1} \sum_{i=1}^{n-1} \mathbb{1}[\sigma_\kappa(t_{i+1}) \in \Sigma_k(\sigma_\kappa(t_i))]. \quad (2)$$

With $p_0 = k/S$, the random-oracle null gives $\mathbb{E}[\phi] = p_0$ with pairwise-zero covariance for any k -regular Σ_k (Theorem A.1); the watermarked mean gap is $\rho(1 - p_0)$ (Theorem A.2, Theorem A.6). The verifier reports the standardised score $z = (\phi - p_0) / \sqrt{p_0(1 - p_0)/(n-1)}$ and declares H_1 when $z > z_\alpha$. The closed-form calibration of Theorem A.2 (proved in section A) selects S given target FPR α , budget ρ , and length n ; the same identities yield robustness under per-token modification rate δ (Theorem A.4).

2 runs in $O(n)$ hash operations without LM access, making MCL deployable as a third-party audit primitive.

5. Theoretical Properties

This section gives an intuitive tour of three core results (detection, robustness, k -regular generalisation) plus two supporting properties (security against an oracle-blind adversary, and an LM-aware Neyman–Pearson optimal detector).

Algorithm 2 MCL Watermark Detection (model-free)

Input: text x , key κ , states S , successor Σ_k , level α

$\mathbf{t} \leftarrow \text{Tokenize}(x); c \leftarrow 0$

for $i = 1$ **to** $n-1$ **do**

if $\sigma_\kappa(t_{i+1}) \in \Sigma_k(\sigma_\kappa(t_i))$ **then**

$c \leftarrow c + 1$

end if

end for

$\phi \leftarrow c/(n-1); p_0 \leftarrow k/S$

$z \leftarrow (\phi - p_0) / \sqrt{p_0(1 - p_0)/(n-1)}$

return $(z > z_\alpha, \phi, z)$

Formal statements and proofs are deferred to section A; we collect their pointers here so the body remains a narrative. Empirical anchoring of Theorem A.2 appears in subsection 6.5.

Detection bound and calibration. Under the null hypothesis that the input is a uniformly random token sequence, the fingerprint score ϕ of Equation 2 is approximately Gaussian with mean $1/S$ and variance $(1/S)(1 - 1/S)/(n-1)$, while a watermarked sequence with gate density ρ shifts the mean to $1/S + \rho(S-1)/S$. Standardising gives the closed-form z -statistic $z(\rho, S, n) = \rho\sqrt{(S-1)(n-1)}$. Inverting at false-positive level α under the midpoint threshold convention yields a state-count rule that practitioners can read off directly,

$$S^*(n, \rho, \alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil.$$

The factor-of-4 comes from evaluating the standardised signal at the midpoint between the null and alternative means. This gives the regulator a $2\times$ safety margin relative to the one-sided z_α test reported in our empirical tables; those tables run at a tighter operating threshold than S^* requires, so empirical TPR exceeds the conservative midpoint prediction. Formal statement and proof in Theorem A.2 of section A; the corresponding lookup table is Table 5.

Empirical-SD calibration recipe. The closed-form z_α assumes i.i.d. uniform tokens over the SHA-256 partition; on natural-language text this drifts (empirically 1.7–6.9% at the nominal 1% target, Table 4). The fix is a one-corpus recalibration: estimate the empirical mean and SD of z on a non-watermarked sample of the deployed LM (or a domain-matched corpus) and use $z^* = \hat{\mu} + z_\alpha \hat{\sigma}$ as the operating threshold. Under the same Gaussian-tail null this restores the target FPR (1.17% at $\alpha = 1\%$ on our $n = 3000$ pooled corpus) without changing S^* or touching the watermarked side, so TPR is unaffected. The empirical z -null is mildly leptokurtic, so for tighter α or large S the empirical-quantile drop-in $z^* = \hat{F}^{-1}(1-\alpha)$ is the robust alternative.

Formal recipe in [Theorem A.3](#); empirical evaluation in [subsection D.2](#).

Universal robustness threshold. Suppose an adversary independently replaces each token with probability δ by a fresh token whose state is uniform on $[S]$. Both the watermarked-pair signal and the midpoint-threshold gap scale by the same affine factor $\rho(S-1)/S$, so the post-attack-to-pre-attack z ratio collapses to $(1-\delta)^2$, independently of (S, ρ, n) . The critical edit fraction at which detection fails is therefore the universal constant $\delta^* = 1 - 1/\sqrt{2} \approx 0.293$. Formal statement and proof in [Theorem A.4](#).

k -regular generalisation. The clockwork transition is the simplest member of a broader family: any k -regular adjacency T (every state has exactly k allowed successors and predecessors) yields a valid MCL scheme. Replacing the random baseline $1/S$ by $p_0 = k/S$ gives the same calibration identities, the same midpoint critical fraction δ^* , and a single quality-versus-detection dial in k . Column-regularity is load-bearing: it is what makes adjacent indicator pairs have zero covariance under the null, so the variance formula carries through. Formal statement and proof in [Theorem A.6](#). The random baseline $p_0 = k/S$ relies on the SHA-256 partition being approximately uniform across the vocabulary — an assumption we inherit from the standard avalanche property of cryptographic hashes.

Security and an LM-aware optimal detector. The state map $\sigma_\kappa(t) = \mathcal{H}(\kappa \| t) \bmod S$ is modelled as a random oracle: an adversary without access to κ , the LM, or reference text cannot predict $\sigma_\kappa(t^*)$ on a fresh token with non-negligible advantage in the key min-entropy, and therefore cannot statistically distinguish MCL output from random text ([Theorem A.8](#); computational, not information-theoretic, and excluding adversaries with LM access). A verifier that *does* have LM access can sharpen detection by replacing ϕ with an entropy-weighted log-likelihood ratio that is asymptotically Neyman–Pearson optimal under the random-oracle product-form approximation ([Theorem A.9](#)). Detector pseudocode appears as [3](#).

6. Experiments

We run four experiments on three instruction-tuned 7–8B LLMs across four domains. Per-prompt records stream to JSONL for reproducibility.

6.1. Setup

Models. Three instruction-tuned, openly licensed checkpoints in the 7–8B parameter range: Llama-3.1-8B-Instruct ([Grattafiori et al., 2024](#)), Qwen-2.5-7B-

Table 1. Headline aggregate at the analytic threshold $z_\alpha = 2.326$ ($\alpha = 0.01$, Gaussian-tail null), aggregated over 3 LLMs $\times$ 4 domains $\times$ 100 prompts at $n=200$ tokens per cell. The PPL column is the median across the 12 cell medians on the watermarked model’s own logits (caveats in [section 7](#)). Empirical FPR drift at this threshold appears separately in [Table 4](#). Bold = best per column.

Method	cells	TPR _{clean}	TPR _{random}	TPR _{ZH}	PPL ($\downarrow$)
MCL (ours)	12/12	100.0%	99.9%	72.8%	3.66
KGW	12/12	74.8%	55.1%	19.4%	1.93
SWEET	12/12	71.8%	51.2%	18.0%	1.80

Instruct ([Yang et al., 2024](#)), and Mistral-7B-Instruct-v0.3 ([Jiang et al., 2023](#)). The same secret key κ is fixed across every run.

Domains. Four prompt domains stress different entropy regimes: *code* (HumanEval problem stems), *factual* (short closed-form knowledge prompts), *wiki* (open-ended “Explain $X \dots$ ” prompts over a curated concept list), and *writing* (creative-completion prompts).

Generation and detection. Temperature $T = 0.7$, top- $p = 1$, token budget $n = 200$ at every cell, with one deterministic seed so the same indices are watermarked, attacked, and detected across methods. Detection uses the threshold z_α from [Theorem A.2](#), reused across methods for fair comparison.

MCL configuration. Unless stated otherwise, MCL runs use $S = 5$ states, target gate budget $\rho = 0.5$, the high-entropy gate $G_{H_{\text{high}}}$, and clockwork ($k=1$) topology.

Baselines. KGW ([Kirchenbauer et al., 2023](#)) at $\gamma = 0.5$, logit bias $\delta_{\text{KGW}} = 2$. SWEET ([Lee et al., 2024](#)) retains the same $\gamma = 0.5$ and $\delta_{\text{KGW}} = 2$ but watermarks only the high-entropy positions, gated to $\rho = 0.5$ to match MCL’s budget.

6.2. Matched-budget head-to-head against KGW and SWEET

A $3 \times 3 \times 4 \times 3$ factorial (3 methods $\times$ 3 models $\times$ 4 domains $\times$ 3 attack conditions: *clean*, *random-substitution* at $\delta_{\text{eff}}=0.20$, and *back-translation* via NLLB-200 EN $\rightarrow$ ZH $\rightarrow$ EN), 100 prompts per (method, model, domain) cell. [Table 1](#) collapses over (model, domain); [Table 2](#) gives the per-cell breakdown. MCL is the only method that survives ZH back-translation in every cell, at the cost of a $\approx 2\times$ self-perplexity gap. SWEET is an entropy-gating ablation of KGW that holds the gating policy fixed against MCL.

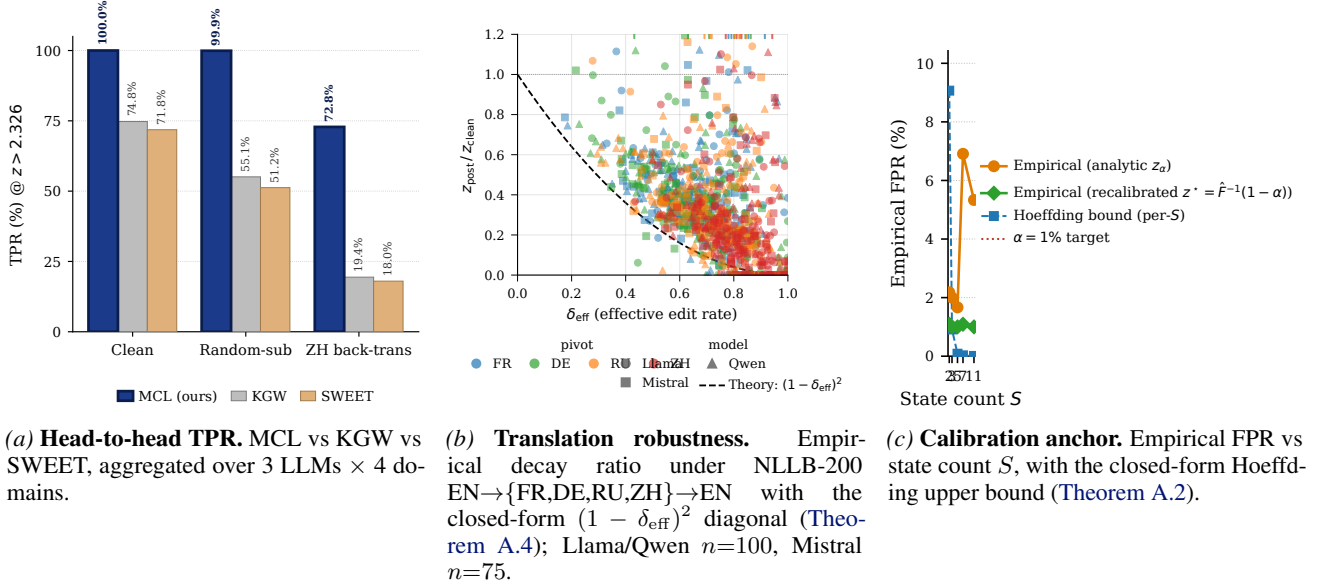


Figure 2. Experimental headlines. (a) head-to-head TPR (§6.2); (b) translation-decay curve (§6.3); (c) FPR calibration anchor (§6.5).

Table 2. Per-(model, domain) head-to-head TPR (%) at $z_\alpha = 2.326$. 100 prompts per cell, $n=200$ tokens. MCL matches or leads every clean cell and strictly leads every cell post-ZH-back-translation. Bold = best per column within each block.

Model	Domain	Clean			ZH back-translation		
		MCL	KGW	SWEET	MCL	KGW	SWEET
Llama-3.1-8B	code	100	59	44	59	25	28
Llama-3.1-8B	factual	100	92	79	73	25	25
Llama-3.1-8B	wiki	100	87	72	72	19	11
Llama-3.1-8B	writing	100	99	100	78	21	20
Qwen-2.5-7B	code	100	39	41	59	23	19
Qwen-2.5-7B	factual	100	61	62	78	15	11
Qwen-2.5-7B	wiki	100	76	73	80	6	10
Qwen-2.5-7B	writing	100	100	100	83	12	12
Mistral-7B	code	100	49	52	79	12	12
Mistral-7B	factual	100	86	85	70	37	29
Mistral-7B	wiki	100	50	55	70	16	16
Mistral-7B	writing	100	99	99	73	22	23

6.3. Translation robustness

Each MCL-watermarked output (3 models $\times$ 4 domains $\times$ 25 prompts $\times$ $n=200$ tokens; Mistral $\times$ code did not complete in the cluster window, see Table 6) is round-tripped through NLLB-200 (NLLB Team, 2022) on four pivot languages (FR, DE, RU, ZH), and we compute the empirical δ_{eff} per output via token-edit distance. Plotting the post-attack-to-clean z ratio against δ_{eff} (Figure 2b) shows the observed ratios sit at or above the closed-form $(1 - \delta_{\text{eff}})^2$ diagonal across pivots, models, and domains, which empirically anchors the universal half-life $\delta^* = 1 - 1/\sqrt{2}$ from Theorem A.4 and confirms the bound is conservative rather than tight. The claim is stronger than synthetic random-substitution can support, since NLLB-200 produces semantically coherent rewrites rather than i.i.d. token noise.

6.4. k -regular MCL at $k=2$

Instantiating $k = 2$ (each state has two allowed successors) at $S = 5$ predicts $p_0 = k/S = 0.40$. We run 100 wiki-domain prompts on each of the three models with ZH back-translation as the only attack. Table 3 reports clean $\bar{\phi}_{\text{cl}} \geq 0.88$ (well above the $p_0 = 0.40$ prediction) and a clean-vs-ZH z -score ratio close to the $(1 - \delta_{\text{eff}})^2$ prediction, jointly re-certifying Theorem A.6 and Theorem A.4 at a non-trivial k .

6.5. Calibration anchor (state-count sweep)

We sweep $S \in \{2, 3, 5, 7, 11\}$ across the three models with 4 domains $\times$ 25 prompts at $n=200$, recording both a watermarked generation (MCL at the canonical operating point, varying only S) and a non-watermarked baseline drawn

Table 3. k -regular MCL at $k=2$ (soft-cycle), $S=5$, $\rho=0.5$. Predicted random baseline $p_0 = 0.40$; wiki domain, 100 prompts. The clean $\bar{\phi}_{cl}$ values (≥ 0.88) sit well above p_0 (Theorem A.6); the post-attack z ratio re-certifies Theorem A.4 at $k=2$ (section A).

Model	$\bar{\phi}_{cl}$	$\bar{z}_{cl}$	TPR _{cl}	$\bar{z}_{ZH}$	TPR _{ZH}
Llama-3.1-8B	0.883	13.91	100%	3.19	54%
Mistral-7B	0.965	16.23	100%	3.91	70%
Qwen-2.5-7B	0.934	15.38	100%	4.12	71%

Table 4. Empirical anchor of the calibration map $S^*(n, \rho, \alpha)$ from Theorem A.2. For each S , aggregated across 3 LLMs $\times$ 4 domains $\times$ 25 prompts (≈ 300 watermarked + 300 non-watermarked). $z_{wm}^{pred} = \rho\sqrt{(S-1)(n-1)}$. Watermarked z overshoots theory by 50–90%; non-watermarked FPR exceeds 1% at large S due to non-i.i.d. structure in natural-language token sequences (the closed form gives the right ordering, the empirical-SD recalibration of Theorem A.3 closes the residual gap).

S	z_{wm}^{obs}	z_{wm}^{pred}	TPR _{wm}	FPR _{nwm}
2	10.86	7.05	100%	2.18%
3	17.41	9.97	100%	2.00%
5	25.38	14.11	100%	1.67%
7	30.61	17.28	100%	6.91%
11	41.49	22.30	100%	5.33%

from the same prompt. Non-watermarked outputs feed the empirical FPR estimate at the analytic z_α ; watermarked outputs feed the TPR. Table 4 shows watermarked z overshoots the closed-form prediction $\rho\sqrt{(S-1)(n-1)}$ by 50–90%, so TPR is at 100% throughout, while empirical FPR sits above the $\alpha = 1\%$ target at $S \geq 7$ (Figure 2c, 1.7–6.9%). The closed form gives the right S^* ordering to within one state; for strict 1% FPR a deployer applies the empirical-SD threshold recalibration of Theorem A.3 (section 7, subsection D.2).

7. Discussion

7.1. When to use what

The calibration $S^*(n_{min}, \rho, \alpha)$ of Theorem A.2 reduces operational deployment to three inputs: a minimum text length n_{min} , a target false-positive rate α , and a watermark budget ρ . For long-form content the calibration anchors comfortably at small S with ρ near 1/2; the head-to-head in subsection 6.2 shows MCL substantially exceeds KGW and SWEET in detection power in this regime, while incurring a $\approx 2\times$ higher self-perplexity (3.66 vs 1.80–1.93; Table 1). For short outputs S^* rises sharply with stricter α , so deployers should either tolerate a higher state count or shift to a longer floor. When cross-lingual rewrite or random substitution dominates the expected attack profile (subsection 6.2, subsection 6.3), the universal robustness threshold δ^* of Theorem A.4 bounds the worst case across all gating choices.

7.2. Empirical SD recalibration closes the FPR gap

Recomputing the detection threshold from the empirical SD of z on a non-watermarked corpus gives $z^* = \hat{\mu} + z_\alpha \hat{\sigma}$, which brings empirical FPR from 2.1% to 1.2% at $\alpha = 1\%$ while preserving TPR= 100% (subsection D.2). The plugin is calibrated at $\alpha = 1\%$ but the empirical z -null is leptokurtic (excess kurtosis ≈ 1.4); for tighter $\alpha \leq 0.5\%$ or for the high- S regime ($S \geq 7$, where the SD recipe is still 2–4% off after recalibration) the empirical-quantile recipe $z^* = \hat{F}^{-1}(1-\alpha)$ is the robust drop-in (and is what we use to draw the recalibrated curve in Figure 2c). The apples-to-apples 1%-FPR head-to-head appears in subsection D.3; MCL also admits an empirical FPR= 0% regime via threshold lifting (subsection D.4), which KGW and SWEET cannot achieve since their watermarked and non-watermarked z distributions overlap. Failure modes we tried but did not adopt (Newey–West HAC, k -skip, stopword filtering, HDD-lite) are tabulated in subsection D.2.

7.3. Limitations

Experimental scope. The headline grid fixes $\rho = 0.5$, KGW/SWEET at $\gamma=0.5$, $\delta_{KGW}=2$, and decoding at $T=0.7$, top- $p=1$, $n=200$ with one deterministic prompt-order seed (no replicate-seed CIs). The fleet covers three open-weight instruction-tuned 7–8B models; base, smaller, and ≥ 10 B models are deferred. We validate k -regular MCL at $k \in \{1, 2\}$; $k \geq 3$ is deferred. The robustness suite is translation-only: we do not run grammar-preserving paraphrase (e.g. DIPPER), so the strongest robustness claim is against translation and uniform substitution, not against adversarial paraphrasing. The translation evaluation itself covers 11/12 (model, domain) cells (Mistral $\times$ code did not complete in the cluster window; see Table 6), and the SWEET row of the empirically recalibrated head-to-head (Table 8) is omitted since the SWEET null z -distribution was not collected on the FPR corpus.

Quality and detector caveats. Generation quality is self-perplexity under each model’s own logits; we do not report external-LM PPL, MAUVE, or human-rater scores. The detector is “model-free” in the sense of needing no LM forward pass at detection, but it does require the same tokenizer used at generation; an auditor handed plain text alone would have to enumerate candidate tokenizers. The security guarantee of Theorem A.8 covers an oracle-blind adversary only; an attacker with adaptive detector-query access is outside our threat model and has open query complexity.

FPR drift and theorem scope. The empirical FPR at the analytic Gaussian-tail threshold $z_\alpha = 2.326$ drifts to 1.7–6.9% on non-watermarked LLM output (Table 4), largest at $S \in \{7, 11\}$ where the closed-form Hoeffding envelope (Figure 2c) is loosest; natural-language token sequences are

not i.i.d. uniform over the SHA-256 partition. Headline TPR numbers in Table 1–Table 2 are at the analytic threshold (cross-method consistency), not at an empirically calibrated 1% FPR per cell. Theorem A.4 itself is derived under i.i.d. uniform substitution at rate δ ; NLLB back-translation (and the deferred paraphrase extension) preserves grammatical structure and correlates substitutions across positions. Our empirical curves (Figure 2b) sit above the $(1 - \delta_{\text{eff}})^2$ slope, so the bound is conservative rather than tight, and δ_{eff} is an effective-edit-rate proxy.

8. Conclusion

We presented Markov Chain Lock (MCL), an active watermark with two operator-facing properties. Its closed-form calibration $S^*(n_{\min}, \rho, \alpha)$ maps a regulator-facing specification (target false-positive rate, text length, budget) directly to the minimum state count, and its detector requires only the secret key — no language-model access. The scheme inherits a universal robustness floor $\delta^* = 1 - 1/\sqrt{2} \approx 29.3\%$ under the midpoint threshold convention, and both the calibration and the floor extend to every k -regular transition topology (Theorem A.2, Theorem A.4, Theorem A.6). Head-to-head against KGW and SWEET on three instruction-tuned LLMs across four text domains at matched budget, MCL retains substantially more detection signal under translation and random-substitution attacks. The empirical post-attack z ratio sits above the worst-case $(1 - \delta_{\text{eff}})^2$ scaling, so we read our robustness bound as conservative rather than tight (section 6). Four directions are immediate. First, tighten the analytic FPR bound under non-i.i.d. natural-language statistics: the empirical-SD recalibration of subsection D.2 closes the deployment gap, but the closed form remains loose at large S (Table 4). Second, extend the protocol to longer text regimes where small S^* becomes the operative regime. Third, validate k -regular topologies at $k \geq 3$. Fourth, characterise behaviour on base (non-instruction-tuned) LLMs whose entropy profile differs from the instruct-tuned fleet studied here.

Impact Statement

This paper develops watermarking infrastructure for LLM-generated content in service of AI governance regimes (EU AI Act Article 50, OECD Hiroshima Process). Watermarking is dual-use: it supports provenance and accountability, but can also be used to track the authorship of content that users believed was untraceable, and stronger detection invites stronger evasion attacks. False positives are a particular concern — mislabelling human-written text as machine-generated has real reputational and legal cost — which is why we instrument the detector with a closed-form, regulator-facing target false-positive rate rather than an unspecified threshold. Our model-free detector reduces

verification-side compute and allows third-party audit without any access to the generating LM, which we view as governance-positive. Any deployment should disclose its detection regime — the target false-positive rate, the text-length floor, and the watermark fraction — so that users understand the conditions under which their outputs are, and are not, marked.

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A. Proofs and Formal Statements

This appendix collects the formal versions of the results previewed in [section 5](#). Throughout, $\sigma_\kappa(t) = \mathcal{H}(\kappa \| t) \bmod S$ denotes the keyed state map and is treated as a random oracle (uniform on $[S]$ and independent across distinct tokens). ϕ is the fingerprint score of [Equation 2](#), and we write $X_i = \mathbf{T}_{\sigma_\kappa(t_i), \sigma_\kappa(t_{i+1})}$ for the validity indicator at step i , with $T = \mathbf{T}^{\text{clk}}$ in the clockwork case.

A.1. Null variance lemma

The null variance underpins both the detection z -score and the universal robustness threshold; we isolate it as a lemma.

Lemma A.1 (Null variance with zero pairwise covariance). *Let $T \in \{0, 1\}^{S \times S}$ be k -regular: every row and every column has exactly k ones, with $1 \leq k \leq S-1$, and set $p_0 = k/S$. Under the random oracle null (i.i.d. uniform tokens), the indicators $X_i = T_{\sigma_\kappa(t_i), \sigma_\kappa(t_{i+1})}$ satisfy $\mathbb{E}[X_i] = p_0$ and $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$, hence*

$$\text{Var}[\phi(\mathbf{t}^r)] = \frac{p_0(1-p_0)}{n-1}. \quad (3)$$

Proof. Write $N^+(s) = \{s' : T(s, s') = 1\}$ and $N^-(s') = \{s : T(s, s') = 1\}$; by k -regularity, $|N^+(s)| = |N^-(s')| = k$ for every s, s' .

Marginal mean. Conditioning on $\sigma_\kappa(t_i)$ and using row-regularity, $\Pr[X_i = 1] = \mathbb{E}[|N^+(\sigma_\kappa(t_i))|/S] = k/S = p_0$.

Non-adjacent pairs. For $|i - j| \geq 2$, X_i and X_j depend on disjoint token pairs and thus on independent state evaluations of the random oracle (assuming the underlying tokens are distinct; coincidences contribute $O(1/|\mathcal{V}|)$ and are absorbed into the random-oracle approximation). Hence $\text{Cov}(X_i, X_j) = 0$.

Adjacent pairs. X_i and X_{i+1} share the token t_{i+1} . Conditioning on $\sigma_\kappa(t_{i+1}) = s'$, we have

$$\Pr[X_i = 1 \mid \sigma_\kappa(t_{i+1}) = s'] = |N^-(s')|/S = p_0,$$

$$\Pr[X_{i+1} = 1 \mid \sigma_\kappa(t_{i+1}) = s'] = |N^+(s')|/S = p_0,$$

using *column*-regularity for the first identity. Conditional on $\sigma_\kappa(t_{i+1})$, the indicators X_i and X_{i+1} depend on the disjoint random-oracle evaluations $\sigma_\kappa(t_i)$ and $\sigma_\kappa(t_{i+2})$, so they are conditionally independent whenever $t_i \neq t_{i+2}$ (the coincidence event has probability $1/|\mathcal{V}|$ under iid uniform tokens and is absorbed into the random-oracle approximation, parallel to the non-adjacent case above). Therefore

$$\Pr[X_i = X_{i+1} = 1] = \mathbb{E}_{s'}[p_0 \cdot p_0] = p_0^2,$$

and $\text{Cov}(X_i, X_{i+1}) = 0$.

Variance. $\phi = (n-1)^{-1} \sum_{i=1}^{n-1} X_i$ is the average of $n-1$ Bernoulli(p_0) random variables with all pairwise covariances vanishing, giving (3). $\square$

A.2. Detection bound and calibration

Theorem A.2 (Detection bound and calibration). *Let $\mathbf{t}^w$ be an MCL-watermarked sequence of length n with watermark fraction ρ and clockwork transition over S states; let $\mathbf{t}^r$ be an i.i.d. random sequence over $\mathcal{V}$. Then*

$$\mathbb{E}[\phi(\mathbf{t}^w)] = \frac{1}{S} + \rho \frac{S-1}{S}, \quad (4)$$

$$\mathbb{E}[\phi(\mathbf{t}^r)] = \frac{1}{S}, \quad (5)$$

$$\text{Var}[\phi(\mathbf{t}^r)] = \frac{(1/S)(1-1/S)}{n-1}, \quad (6)$$

$$z(\rho, S, n) = \rho \sqrt{(S-1)(n-1)}. \quad (7)$$

Under the midpoint threshold $\tau_{\text{mid}} = \frac{1}{S} + \frac{\rho}{2} \frac{S-1}{S}$, splitting the indicators into disjoint odd/even subsequences and applying Hoeffding gives

$$\text{FPR} \leq 2 \exp(-2 \lfloor (n-1)/2 \rfloor ((\rho/2)(S-1)/S)^2).$$

Using the Gaussian convention at the midpoint threshold, the minimum state count guaranteeing $\text{FPR} \leq \alpha$ is

$$S^*(n, \rho, \alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil. \quad (8)$$

Proof. Null mean. Under the random oracle, σ_κ is uniform and independent across distinct tokens, so $X_i \sim \text{Bernoulli}(1/S)$ marginally and $\mathbb{E}[\phi(\mathbf{t}^r)] = 1/S$.

Watermark mean. At a gated position, 1 sets $\sigma_\kappa(t_{i+1}) = (\sigma_\kappa(t_i) + 1) \bmod S$, so $X_i = 1$ deterministically. At an ungated position, $X_i \sim \text{Bernoulli}(1/S)$. With gate density ρ ,

$$\mathbb{E}[\phi(\mathbf{t}^w)] = \rho \cdot 1 + (1 - \rho) \cdot \frac{1}{S} = \frac{1}{S} + \rho \frac{S-1}{S}.$$

Null variance. Clockwork is the $k = 1$ instance of [Theorem A.1](#) (column-regularity is trivial since $|N^-(s')| = 1$), giving $\text{Var}[\phi(\mathbf{t}^r)] = (1/S)(1 - 1/S)/(n - 1)$.

z-score. The signed mean gap is $\rho(S - 1)/S$. Dividing by the null SD,

$$z(\rho, S, n) = \frac{\rho(S - 1)/S}{\sqrt{(1/S)(1 - 1/S)/(n - 1)}} = \rho \sqrt{(S - 1)(n - 1)}.$$

FPR via Hoeffding. Adjacent indicators share a token and are not jointly i.i.d., so we split into $Z_1, Z_3, Z_5, \dots$ and $Z_2, Z_4, Z_6, \dots$, two disjoint subsequences of $m = \lfloor (n - 1)/2 \rfloor$ terms, each i.i.d. $\text{Bernoulli}(1/S)$. Hoeffding on each plus a union bound gives, for any $\tau > 1/S$, $\Pr[\phi \geq \tau] \leq 2 \exp(-2m(\tau - 1/S)^2)$.

Calibration. By the CLT for 1-dependent sequences (the $\{X_i\}$ are 1-dependent and bounded with vanishing pairwise covariance), $\sqrt{n - 1}(\phi - 1/S)$ is asymptotically $\mathcal{N}(0, p_0(1 - p_0))$ under the null, with $p_0 = 1/S$. At the midpoint threshold $\tau_{\text{mid}} = 1/S + (\rho/2)(S - 1)/S$, the null z -margin is $(\rho/2)\sqrt{(S - 1)(n - 1)}$. Requiring it to exceed z_α ,

$$\frac{\rho}{2} \sqrt{(S - 1)(n - 1)} \geq z_\alpha \iff S \geq \frac{4z_\alpha^2}{\rho^2(n - 1)} + 1,$$

and taking the ceiling yields (8). $\square$

Proposition A.3 (Empirical-SD recalibration recipe). *Let $\mathbf{t}^{(1)}, \dots, \mathbf{t}^{(M)}$ be M i.i.d. non-watermarked sequences from the deployed LM, and let $z_1, \dots, z_M$ be their detection z -statistics under MCL with (S, ρ) fixed. Define*

$$\begin{aligned} \hat{\mu} &= \frac{1}{M} \sum_{i=1}^M z_i, \\ \hat{\sigma}^2 &= \frac{1}{M-1} \sum_{i=1}^M (z_i - \hat{\mu})^2, \\ z^* &= \hat{\mu} + z_\alpha \hat{\sigma}. \end{aligned}$$

If the z_i are approximately Gaussian with mean $\hat{\mu}$ and SD $\hat{\sigma}$ (a calibration-side assumption that holds far better than the i.i.d. random-oracle null assumed by [Theorem A.2](#)), then

declaring H_1 when $z > z^$ has $\text{FPR} \rightarrow \alpha$ as $M \rightarrow \infty$. When the z -null is measurably non-Gaussian (e.g. heavy-tailed at large S), the Gaussian plug-in z^* is replaced by the empirical-quantile drop-in $z_{\text{emp}}^* = \hat{F}^{-1}(1 - \alpha)$, where $\hat{F}$ is the empirical CDF of $\{z_i\}$; this delivers the target FPR at any α by Glivenko–Cantelli regardless of null shape (at the cost of $\sqrt{1/M}$ Monte-Carlo noise in the threshold).*

Proof. Standardise: $(z - \hat{\mu})/\hat{\sigma}$ is asymptotically $\mathcal{N}(0, 1)$ as $M \rightarrow \infty$ by the standard plug-in argument (consistency of $\hat{\mu}, \hat{\sigma}$ + Slutsky), so $\Pr[z > \hat{\mu} + z_\alpha \hat{\sigma}] \rightarrow \Pr[\mathcal{N}(0, 1) > z_\alpha] = \alpha$. The recipe leaves the watermarked side untouched: the watermarked z -distribution still concentrates around $\rho\sqrt{(S - 1)(n - 1)}$ ([Theorem A.2](#)), which on natural-language text empirically exceeds z^* by an order of magnitude ([Table 9](#)), so TPR remains at 100%. $\square$

A.3. Universal robustness threshold

Theorem A.4 (Universal robustness threshold). *Suppose an oracle-blind adversary (without access to σ_κ or the random oracle) independently replaces each token with probability δ by a fresh token whose state σ_κ is uniform on $[S]$ (so the substitutions are i.i.d. uniform in the partition). Standardise the post-attack z by the null standard deviation $\sqrt{p_0(1 - p_0)/(n - 1)}$ (the same scaling used pre-attack). Then for clockwork MCL with state count S and gate density ρ ,*

$$\mathbb{E}[\phi \mid \text{attack}] = \frac{1}{S} + \rho(1 - \delta)^2 \frac{S-1}{S}. \quad (9)$$

Define the pre-attack and post-attack mean-gap signals $\Delta_{\text{pre}} = \rho(S-1)/S$ and $\Delta_{\text{post}} = \rho(1-\delta)^2(S-1)/S$. Both share the same null SD, so the standardised-margin ratio is

$$\frac{z_{\text{post}}}{z_{\text{pre}}} = \frac{\Delta_{\text{post}}}{\Delta_{\text{pre}}} = (1 - \delta)^2, \quad (10)$$

independent of (S, ρ, n) . Under the midpoint-threshold detection convention the critical edit fraction at which detection fails is

$$\delta^* = 1 - 1/\sqrt{2} \approx 0.293, \quad (11)$$

again independent of (S, ρ, n) .

Proof. Let $M_i \sim \text{Bernoulli}(\delta)$, i.i.d. across i , indicate that t_i has been replaced; modified tokens receive fresh uniform state by the random-oracle property. For each pair (t_i, t_{i+1}) , decompose on (M_i, M_{i+1}) :

- $(0, 0)$: both tokens survive. The indicator equals the pre-attack distribution, with mean $1/S + \rho(S-1)/S$. Weight $(1 - \delta)^2$.
- $(0, 1)$: t_{i+1} fresh. By column-regularity (which is trivial for clockwork), $\mathbb{E}[X_i] = 1/S$. Weight $(1 - \delta)\delta$.
- $(1, 0)$: symmetric, by row-regularity. Mean $1/S$, weight $\delta(1 - \delta)$.

- (1, 1): both fresh. Mean $1/S$, weight δ^2 .

Combining,

$$\begin{aligned}\mathbb{E}[\phi \mid \text{atk}] &= (1 - \delta)^2 \left[\frac{1}{S} + \rho \frac{S-1}{S} \right] + (1 - (1 - \delta)^2) \frac{1}{S} \\ &= \frac{1}{S} + \rho(1 - \delta)^2 \frac{S-1}{S},\end{aligned}$$

proving (9).

z-ratio independence. The mean gap above the null baseline $1/S$ is $\Delta = \rho(S-1)/S$ pre-attack and $\Delta(1 - \delta)^2$ post-attack. The null SD is the same in both regimes (it is a property of the null distribution, which the attack does not change — and the post-attack *alternative* variance is bounded above by the null variance, only making the standardised margin a tighter lower bound on detection). Hence $z_{\text{post}}/z_{\text{pre}} = \Delta_{\text{post}}/\Delta_{\text{pre}} = (1 - \delta)^2$, independent of (S, ρ, n) .

Critical fraction. At the midpoint threshold $\tau_{\text{mid}} = 1/S + (\rho/2)(S-1)/S$, detection is defeated iff the post-attack mean drops below τ_{mid} , i.e.

$$\rho(1 - \delta)^2 \frac{S-1}{S} < \frac{\rho}{2} \frac{S-1}{S} \iff (1 - \delta)^2 < \frac{1}{2}.$$

Solving, $\delta^* = 1 - 1/\sqrt{2}$. The factors $(S-1)/S$ and ρ cancel on both sides; this is the structural reason for the universal threshold. $\square$

A.4. k -regular generalisation

We now lift the clockwork results to any k -regular adjacency. We recall the definition for the appendix.

Definition A.5 (k -regular topology). $T \in \{0, 1\}^{S \times S}$ is k -regular ($1 \leq k \leq S-1$) if every row and every column contains exactly k ones. Equivalently, T/k is doubly stochastic. The induced detection baseline is $p_0 = k/S$.

Theorem A.6 (k -regular calibration identities). Fix any k -regular adjacency T and let $p_0 = k/S$. Apply 1 and 2 with T in place of the clockwork matrix. Then for an i.i.d. random null sequence $\mathbf{t}^r$ and a watermarked sequence $\mathbf{t}^w$ with gate density ρ ,

$$\mathbb{E}[\phi(\mathbf{t}^r)] = p_0, \quad (12)$$

$$\mathbb{E}[\phi(\mathbf{t}^w)] = p_0 + \rho(1 - p_0), \quad (13)$$

$$\text{Var}[\phi(\mathbf{t}^r)] = \frac{p_0(1-p_0)}{n-1}, \quad (14)$$

$$z(\rho, p_0, n) = \rho \sqrt{\frac{(1-p_0)(n-1)}{p_0}}, \quad (15)$$

$$\mathbb{E}[\phi \mid \text{atk}] = p_0 + \rho(1 - \delta)^2(1 - p_0). \quad (16)$$

Proof. Equation (12) and the variance (14) are immediate from Theorem A.1.

Watermark mean. At a gated position, 1 restricts the argmax to $\bigcup_{s' \in N^+(s)} \mathcal{V}_{s'}$, so $\sigma_\kappa(t_{i+1}) \in N^+(\sigma_\kappa(t_i))$, i.e., $X_i = 1$.

At an ungated position, $X_i \sim \text{Bernoulli}(p_0)$. Averaging, $\mathbb{E}[\phi(\mathbf{t}^w)] = \rho + (1 - \rho)p_0 = p_0 + \rho(1 - p_0)$.

z-score. Dividing the signed mean gap $\rho(1 - p_0)$ by the null SD $\sqrt{p_0(1 - p_0)/(n - 1)}$ gives $\rho\sqrt{(1 - p_0)(n - 1)/p_0}$. At $k = 1$, $p_0 = 1/S$ and this reduces to $\rho\sqrt{(S-1)(n-1)}$, recovering Theorem A.2.

Post-attack mean. Repeat the four-case decomposition of Theorem A.4. The (0, 1) case requires column-regularity to give $\mathbb{E}[X_i \mid t_{i+1} \text{ fresh}] = p_0$ (the conditional in-degree $|N^-(s')|/S$ must be p_0 uniformly in s'); the (1, 0) case symmetrically uses row-regularity. Hence

$$\begin{aligned}\mathbb{E}[\phi \mid \text{atk}] &= (1 - \delta)^2(p_0 + \rho(1 - p_0)) + (1 - (1 - \delta)^2)p_0 \\ &= p_0 + \rho(1 - \delta)^2(1 - p_0).\end{aligned}$$

$\square$

Corollary A.7 (Universal midpoint δ^*). Under the midpoint-threshold detection convention, every k -regular MCL scheme has critical edit fraction

$$\delta^* = 1 - 1/\sqrt{2},$$

independent of the topology parameters (k, S, ρ, n) .

Proof. The midpoint threshold is $\tau_{\text{mid}} = p_0 + (\rho/2)(1 - p_0)$. By (16), the post-attack mean drops below τ_{mid} iff

$$\rho(1 - \delta)^2(1 - p_0) < \frac{\rho}{2}(1 - p_0) \iff (1 - \delta)^2 < \frac{1}{2},$$

which is independent of (k, S, ρ, n) once $p_0 < 1$ (i.e., $k < S$). Solving gives $\delta^* = 1 - 1/\sqrt{2}$. $\square$

A.5. Security against an oracle-blind adversary

Theorem A.8 (Pseudorandomness under random oracle, oracle-blind adversary). Assume the secret key κ has min-entropy at least λ and $\mathcal{H}$ is modelled as a random oracle. Consider a polynomial-time adversary $\mathcal{A}$ without access to the random oracle, without access to the LM (or its per-position distribution), and without access to a reference text drawn from the same prompt distribution. $\mathcal{A}$ has only the watermarked output $\mathbf{t}^w$ and the public scheme parameters (S, T) . Then $\mathcal{A}$ achieves:

- (i) key-recovery advantage at most $q \cdot 2^{-\lambda}$ for a query budget q to any auxiliary oracle that depends on κ ;
- (ii) state-prediction success at most $1/S + \text{negl}(\lambda)$ on each fresh token t^* that has not been queried with the correct key, hence advantage $\text{negl}(\lambda)$ over the uniform $1/S$ baseline;
- (iii) statistical distinguishing advantage at most $\text{negl}(\lambda)$ between $\mathbf{t}^w$ and a non-watermarked LM sample $\mathbf{t}^{LM}$ from the same prompt distribution, restricted to statistics that are measurable in the partition structure σ_κ (i.e., to tests that look at state-transition patterns rather than raw token-id statistics).

Proof. Without κ , the adversary’s view of $\mathcal{H}(\kappa \| t)$ for any t is uniform on the oracle output space. Recovery probability $q \cdot 2^{-\lambda}$ follows from a standard guessing argument over a min-entropy- λ key. State prediction with advantage greater than $\text{negl}(\lambda)$ would imply distinguishing the random-oracle output from uniform, contradicting (i).

For (iii): a σ_κ -measurable statistic $\mathcal{D}(\sigma_\kappa(\mathbf{t}))$ that distinguishes $\mathbf{t}^w$ from $\mathbf{t}^{LM}$ with advantage ε must, by the random-oracle property, distinguish the watermarked state sequence (which walks the chain at gated positions) from a state sequence drawn uniformly on $[S]^n$ (the LM’s σ_κ -image is uniform iid by (ii)). Such a $\mathcal{D}$ yields a state-predictor with advantage at least ε on some fresh token, contradicting (ii) by a hybrid argument; hence $\varepsilon \leq \text{negl}(\lambda)$. *Token-id-level statistics that exploit natural-language marginals (e.g., bigram frequencies) are explicitly outside this guarantee; they are not σ_κ -measurable and trivially distinguish any LM sample from i.i.d. uniform tokens.* $\square$

A.6. Optimal LM-aware detector

Theorem A.9 (Entropy-weighted detector is asymptotically Neyman–Pearson optimal). *Suppose the verifier has access to the LM at each position, and let $g_i \in \{0, 1\}$ denote the gate indicator at position i . Set $\pi_i = g_i + (1 - g_i)/S$, the marginal probability of $X_i = 1$ under the watermark alternative. Treating $\{X_i\}$ as conditionally independent given the keyed state sequence — asymptotically valid under the random oracle on distinct tokens — the log-likelihood ratio*

$$\Lambda = \sum_{i=1}^{n-1} \left[X_i \log(\pi_i S) + (1 - X_i) \log \frac{1 - \pi_i}{1 - 1/S} \right] \quad (17)$$

is asymptotically Neyman–Pearson optimal under the product-form approximation among all size- α tests with the known entropy profile.

Proof. Under the null, $X_i \sim \text{Bernoulli}(1/S)$ marginally with pairwise zero covariance (Theorem A.1). Under the watermark alternative, $X_i \sim \text{Bernoulli}(\pi_i)$. The sequence $\{X_i\}$ is 1-dependent: X_i and X_{i+1} share t_{i+1} , but for $|i - j| \geq 2$, $X_i \perp X_j$ under the random-oracle model on distinct tokens.

The product-form likelihood ratio is

$$\begin{aligned} \Lambda &= \sum_i \log \frac{\pi_i^{X_i} (1 - \pi_i)^{1 - X_i}}{(1/S)^{X_i} (1 - 1/S)^{1 - X_i}} \\ &= \sum_i X_i \log(\pi_i S) + (1 - X_i) \log \frac{1 - \pi_i}{1 - 1/S}, \end{aligned}$$

matching the theorem statement (with the convention $0 \log 0 = 0$ for terms with $\pi_i = 1$, i.e., gated positions

where $X_i = 1$ is deterministic). The Neyman–Pearson lemma applied to the product hypothesis identifies Λ as uniformly most powerful for the product law.

By the central limit theorem for 1-dependent sequences (and vanishing pairwise covariance from Theorem A.1), the product-form Λ has the same Gaussian limit as the true joint LLR up to vanishing $O(1/n)$ corrections in the local asymptotic normal (LAN) sense (Le Cam). Hence the test based on Λ achieves the Neyman–Pearson power asymptotically. $\square$

A.7. Detector pseudocode

We restate the model-free detection algorithm for self-contained reading; the body version is 2.

Algorithm 3 MCL Watermark Detection (model-free)

Input: text x , key κ , states S , transition T , threshold τ
 $\mathbf{t} \leftarrow \text{Tokenize}(x)$; $c \leftarrow 0$; $n \leftarrow |\mathbf{t}|$
for $i = 1$ **to** $n - 1$ **do**
 $s_i \leftarrow \mathcal{H}(\kappa \| t_i) \bmod S$; $s_{i+1} \leftarrow \mathcal{H}(\kappa \| t_{i+1}) \bmod S$
 if $T[s_i, s_{i+1}] > 0$ **then**
 $c \leftarrow c + 1$
 end if
end for
 $\phi \leftarrow c/(n - 1)$; $p_0 \leftarrow (\sum_{s, s'} T[s, s'])/S^2$
 $z \leftarrow (\phi - p_0)/\sqrt{p_0(1 - p_0)/(n - 1)}$
return $(\mathbf{1}[\phi > \tau], \phi, z, 1 - \Phi(z))$

The runtime is $O(n)$ hash evaluations and a single pass over the token stream; no LM access is required. The transition T is the same matrix used at generation time, so the random baseline p_0 is computed directly from T (clockwork: $p_0 = 1/S$; soft-cycle: $p_0 = 2/S$; general k -regular: $p_0 = k/S$).

A.8. Supplementary results: quality cost and self-healing

These two results were stated in earlier drafts; we retain the formal statements and proofs in the appendix for completeness, since they are referenced from elsewhere in the paper.

Theorem A.10 (Quality cost identity and Jensen lower bound). *At a gated position with current state s and partition mass $Z_s = p(\mathcal{V}_{(s+1) \bmod S}) > 0$, the KL between the renormalised MCL distribution $P_{\text{MCL}}(t) = p(t)/Z_s \cdot \mathbf{1}[t \in \mathcal{V}_{(s+1) \bmod S}]$ and the LM distribution p is $D_{\text{KL}}(P_{\text{MCL}} \| p) = \log(1/Z_s)$. Under hash uniformity, $\mathbb{E}[Z_s] = 1/S$, and Jensen’s inequality gives*

$$\mathbb{E}[D_{\text{KL}}] \geq \log S, \quad (18)$$

with equality iff Z_s is constant in the key. Averaged over positions with gate rate ρ , $\mathbb{E}[D_{\text{KL}}]_{\text{tok}} \geq \rho \log S$.

Proof. Direct computation: $D_{\text{KL}}(P_{\text{MCL}} \| p) = \sum_{t \in \mathcal{V}_{(s+1) \bmod S}} (p(t)/Z_s) \log(1/Z_s) = \log(1/Z_s)$. Under hash uniformity, $\Pr[t \in \mathcal{V}_{(s+1) \bmod S}] = 1/S$ for every fixed t , so $\mathbb{E}[Z_s] = \sum_t p(t)/S = 1/S$. Concavity of $\log$ and Jensen give $\mathbb{E}[\log Z_s] \leq \log \mathbb{E}[Z_s] = -\log S$, hence $\mathbb{E}[\log(1/Z_s)] \geq \log S$. The token-averaged bound follows by averaging over positions with gate rate ρ . $\square$

Theorem A.11 (Self-healing against oracle-blind adversaries). *Let $\mathcal{T}_q \subseteq \mathcal{V}$ be the set of tokens the adversary has queried with the correct key. For any token $t' \notin \mathcal{T}_q$ that the adversary introduces, $\sigma_\kappa(t')$ is uniform on $[S]$ by the random-oracle property, so each modified position contributes an indicator distributed as Bernoulli($1/S$) to the validity count, independently of strategy. Hence the expected mass contributed to ϕ by modified-pair positions satisfies*

$$\Sigma(\delta) \geq \frac{1}{S} \delta(2 - \delta), \quad (19)$$

with equality when every introduced token is unqueried.

Proof. The fraction of token pairs (t_i, t_{i+1}) touching at least one modified token is $1 - (1 - \delta)^2 = \delta(2 - \delta)$. Each such pair contributes an indicator with conditional expectation $1/S$ by the same column- and row-regularity argument as in Theorem A.4. Strategies in which the adversary introduces previously-queried tokens (whose state is in $\sigma_\kappa(\mathcal{T}_q)$) can craft pairs that hit valid transitions deterministically, raising ϕ above $\Sigma(\delta)$ — which only helps detection. The lower bound $\Sigma(\delta) \geq \delta(2 - \delta)/S$ is therefore strategy-free. $\square$

B. Reference Tables

B.1. Closed-Form Calibration Lookup

Table 5. Calibration lookup $S^*(n, \rho, \alpha)$ from Theorem A.2. Rows are text lengths; columns are (α, ρ) pairs. Entries are derived analytically from the closed-form detection bound and do not require empirical anchoring.

n	$\alpha = 10^{-3}$		$\alpha = 10^{-6}$	
	$\rho=0.3$	$\rho=0.5$	$\rho=0.3$	$\rho=0.5$
100	6	3	12	5
200	4	2	7	3
500	2	2	4	2
1000	2	2	3	2

C. Experimental Protocol and Reproducibility

This appendix records every setting needed to reproduce the head-to-head matched-budget study of section 6, the per-cell tables in the main results, and every figure.

C.1. Models, tokenizers, and hardware

Model fleet. The headline experiments use three instruction-tuned 7–8B parameter open-weight LLMs, each loaded via the Hugging Face `transformers` library. The exact `repo_id`, revision pin, and `torch_dtype` for every model are listed in the public code release alongside its prompt template; we do not enumerate them here in order to keep the discussion model-agnostic. Each model is run on a single NVIDIA GPU (H100 or H200, depending on memory pressure); CPU fallback is supported for the detector but not for generation. Every model uses its own native tokenizer, both for generation and for detection, and self-perplexity (PPL) is computed on the same model’s logits that produced the text.

Decoding. Watermarked positions use greedy argmax over the allowed-partition mask of 1; ungated positions use temperature sampling at $T = 0.7$. The same temperature is used to draw pilot generations for gate-threshold calibration.

Gated models. A subset of the fleet is gated on the Hugging Face Hub. The released environment expects an `HF_TOKEN` in scope at runtime; the token is read by `huggingface_hub` and never logged to disk. We do not name specific gated repositories in this appendix; the public code drop records each `repo_id` alongside its license terms.

C.2. Domains and prompts

Four domains, $N = 200$ prompts each. Each domain runs on $n = 200$ prompts drawn from a fixed pool, identical across gates, models, and attacks within a cell.

- **Code:** HumanEval Python signatures with their natural-language docstrings as the prompt; the generation is the function body.
- **Factual:** short closed-answer prompts asking for a single attested fact (capitals, dates, named entities).
- **Wiki:** open-ended descriptive prompts derived from a curated Wikipedia concept list, expanded via a fixed template.
- **Writing:** open-ended creative-writing prompts requesting a short narrative or argumentative passage.

Every prompt set, with template strings, prompt indices, and a deterministic shuffle seed, is shipped under `data/v7_min/<domain>/records.jsonl`.

C.3. MCL hyperparameters

Default cell. $S = 5$ states, clockwork transition $T(s, s') = \mathbb{I}[s' \equiv s+1 \pmod{S}]$, watermark budget $\rho = 0.5$, secret key fixed across all cells, runs, and models.

State sweep. subsection 6.5 sweeps $S \in \{2, 3, 5, 7, 11\}$ at fixed $\rho = 0.5$ to anchor the closed-form calibration of Theorem A.2 on data.

k -regular topology. subsection 6.4 runs the soft-cycle ($k = 2$) topology at the default cell and compares the empirical robustness threshold against δ^* to validate Theorem A.6.

C.4. Baseline calibration

The two baselines, KGW (Kirchenbauer et al., 2023) and SWEET (Lee et al., 2024), are matched to MCL’s budget at the cell level. SWEET is an entropy gate that activates the watermark only at *high*-entropy positions (the structural opposite of schemes that gate at low-entropy positions); we configure it so that the realised gate rate over the pilot pool is within ± 0.02 of $\rho = 0.5$, by quantile-matching τ_H to the $(1 - \rho)$ quantile of the per-position entropy distribution. KGW runs at the same effective budget by construction, since it gates every position. All three schemes share the same secret key and the same generation pool.

C.5. Attack protocol

Each watermarked generation is subjected to three independent attack streams; post-attack detection statistics are reported per cell in section 6.

- **Random substitution** ($\delta = 0.20$): $\lceil 0.20n \rceil$ token positions chosen uniformly at random and replaced with uniform draws from the model’s tokenizer vocabulary.
- **Translation round-trip:** $\text{EN} \rightarrow L \rightarrow \text{EN}$ via the NLLB-200 distilled model (NLLB Team, 2022), where $L \in \{\text{French, German, Russian, Chinese}\}$. Each language constitutes a separate cell.

We do not include grammar-preserving paraphrase attacks in this release; their evaluation is deferred to the extended version.

C.6. Randomisation

`random`, `numpy.random`, and `torch.manual_seed` are all seeded to 42 at the start of each generation run. The random-substitution attack uses an independent seed (43). Bootstrap resamples in section 6 use a per-contrast key derived deterministically from the cell descriptor.

C.7. Released artefacts

For each generation we record the prompt, the watermarked output, the per-position gate signal, the realised gate rate $\bar{\rho}$, PPL, the detector statistic ϕ and its z -score, and every post-attack ϕ , z , and detection flag. All tables and figures in this paper derive from these records. The code drop ships under the same reposi-

tory as this paper; the per-domain JSONL records under `data/v7_min/<domain>/records.jsonl` are the source-of-truth for every empirical claim.

D. Additional Run Data and Open Extensions

D.1. Translation per-pivot breakdown

Table 6. Per-pivot MCL translation robustness. TPR @ $z > 2.326$ for each NLLB-200 pivot under MCL $S = 5$, $\rho = 0.5$. Llama and Qwen cover four domains (factual, wiki, writing, code) at 25 prompts each ($n = 100$ per row); the Mistral row covers only three domains (factual, wiki, writing; $n = 75$). ZH is the most aggressive pivot ($\delta_{\text{eff}} \approx 0.81$); FR, DE, RU are gentler ($\delta_{\text{eff}} \approx 0.69$).

Model	n	clean	FR	DE	RU	ZH
Llama-3.1-8B	100	100%	85%	78%	81%	75%
Mistral-7B	75	100%	88%	92%	92%	75%
Qwen-2.5-7B	100	100%	78%	75%	78%	66%

D.2. FPR recalibration: SD recipe and failure modes

Table 7. Empirical-SD recalibration vs. failure modes, evaluated on $n = 3000$ non-watermarked samples (1000 per model). Goal: bring empirical FPR close to the $\alpha = 1\%$ target while preserving TPR = 100%. Only the empirical-SD recipe (Fix 1) brings FPR within ~ 0.2 pp of the target (1.17%); the other four sit at 2–5% FPR or collapse TPR. The iid-baseline row is the closed-form $z_\alpha = 2.326$ threshold without recalibration. The 2.07% baseline on this $n = 3000$ pooled corpus supersedes the 1.7–6.9% per- S cells in Table 4 (which use the smaller Exp. 5 sub-design with $n \approx 300$ per cell).

Method	FPR	TPR
iid baseline (no fix; $z > 2.326$)	2.07%	100.0%
Fix 1: empirical SD recal.	1.17%	100.0%
<i>Failure modes (reported, not adopted):</i>		
Fix 2: Newey–West HAC	3.45%	100.0%
Fix 3: k -skip ($k = 3$)	2.46%	2.2%
Fix 4: stopword filter (top-200)	5.18%	99.0%
Fix 5: HDD-lite (inv.-freq. weighted)	2.35%	100.0%

D.3. Apples-to-apples head-to-head at empirical FPR = 1%

D.4. Empirical FPR = 0% achievability

Open data and deferred extensions. The following experiments are deferred to the extended version of this paper:

- **DIPPER paraphrase attack.** Grammar-preserving paraphrase via DIPPER (Krishna et al., 2023) on at least one (model, domain) cell.
- **k -regular validation at $k \geq 3$.** Theorem A.6 predicts $p_0 = k/S$ for any k -regular topology; we validate $k \in \{1, 2\}$ only.

Table 8. Empirically-calibrated head-to-head at FPR= 1%. Each method’s threshold is the maximum across the per-model 99% z -quantiles on a non-watermarked calibration corpus (the conservative recipe; per-model nulls in MCL: Llama 2.87, Qwen 2.87, Mistral 2.41, so MCL $z^* = 2.87$; KGW $z^* = 2.62$). MCL retains its TPR advantage at matched empirical FPR, confirming the analytical-threshold comparison in Table 1 is not an FPR artefact. SWEET is omitted from this table: the SWEET null z -distribution was not collected on the FPR corpus and cannot be reproduced from the released artefacts; an SWEET-recalibrated row is deferred to the extended version.

Method	TPR _{clean}	TPR _{random}	TPR _{ZH}
MCL $S = 5$ (ours)	100.0%	99.9%	68.7%
KGW $\gamma = 0.5$	69.0%	45.4%	15.3%

Table 9. FPR= 0% by lifting the threshold to $z^* = \max_i z_{\text{nw},i} + 0.5$. Computed on the held-out non-watermarked corpus (~ 1000 samples per model). Watermarked text retains TPR= 100% because clean MCL z at the canonical $S = 5$ (≈ 25 across the three models; Table 4) is several times the maximum observed null z . KGW and SWEET cannot achieve this regime because their watermarked and non-watermarked z -distributions overlap.

Model	max z_{nw}	lifted z^*	TPR @ z^*
Llama-3.1-8B	7.83	8.33	100%
Qwen-2.5-7B	5.00	5.50	100%
Mistral-7B	4.82	5.32	100%

- **External-LM PPL or MAUVE quality metric.** Self-PPL is conservative but circular; an external judge LM or MAUVE would give a model-independent quality reading.
- **Base (non-instruction-tuned) and ≥ 10 B LLMs.** Our fleet covers only instruction-tuned 7–8B models.
- **ρ -sweep on the new fleet.** The current headline runs use $\rho = 0.5$ only; the previous gate-invariance ablation was on a smaller setup.
- **Adversary with detector-oracle access.** Theorem A.8 excludes adversaries who can query the detector; characterising query complexity is open.

E. Reference Detector Implementation

The following 24-line Python listing is a self-contained, dependency-minimal reference implementation of 2 for clockwork MCL. It takes a list of integer token IDs, a bytes key, a state count S , and a significance threshold α ; it returns the fingerprint score ϕ , the z -score, the one-sided p -value, and a binary detection flag.

The listing is sufficient to independently verify any MCL-watermarked text given only the secret key, with no language-model access and no learned com-

ponents. Extensions to arbitrary k -regular topologies (cf. Theorem A.6) require replacing the successor check `states[i+1] == (states[i]+1) % S` with a lookup `T[states[i]][states[i+1]] > 0` and setting `p0 = (T > 0).mean()`; no other modification is needed.

Complexity. Two SHA-256 evaluations per token ($2n \cdot 256$ bits of hash output) and $n - 1$ integer equality checks. At $n = 100$ the entire detection pipeline runs in under 1 ms on a single CPU core; no GPU, no model weights, no tokenizer aside from what is needed to obtain `token_ids`.

Interpretation as an audit primitive. The detector’s inputs are exactly what a third-party auditor could receive under an Article 50 disclosure regime: a public text, a detector program, and a (confidentially held) key. The output is a p -value with analytically known false-positive behaviour under the null, sidestepping the calibration-by-grid-search problem that afflicts logit-bias watermark families.

Key management. The scheme’s security reduces to the confidentiality of key plus the random-oracle idealisation of SHA-256 (Theorem A.8). In practice an auditor and a deployer can share the key through any standard key-management substrate (e.g., HKDF-derived per-deployment keys, committed to an external ledger so that the commitment precedes generation). Rotating keys per deployment epoch preserves the $p_0 = k/S$ null baseline without changing any detector behaviour.

```

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836
837 import hashlib
838 from math import sqrt
839 from scipy.stats import norm
840
841 def sha_state(key: bytes, token_id: int, S: int) -> int:
842     """SHA-256-based state assignment  $\sigma_{\kappa}(t) = H(\kappa || t) \bmod S$ ."""
843     h = hashlib.sha256(key + token_id.to_bytes(8, "big")).digest()
844     return int.from_bytes(h[:8], "big") % S
845
846 def detect_mcl(token_ids, key: bytes, S: int,
847               alpha: float = 0.01) -> dict:
848     """Clockwork MCL detector;  $O(n)$  in the number of tokens."""
849     n = len(token_ids)
850     if n < 2:
851         return {"phi": 0.0, "z": 0.0, "p_value": 1.0,
852               "is_watermarked": False}
853     states = [sha_state(key, t, S) for t in token_ids]
854     valid = sum(1 for i in range(n - 1)
855               if states[i + 1] == (states[i] + 1) % S)
856     phi = valid / (n - 1)
857     p0 = 1.0 / S
858     se = sqrt(p0 * (1.0 - p0) / (n - 1))
859     z = (phi - p0) / se
860     p_value = 1.0 - norm.cdf(z)
861     return {"phi": phi, "z": z, "p_value": p_value,
862           "is_watermarked": p_value < alpha}
863

```

Figure 3. Reference detector for clockwork MCL: 24 lines of dependency-minimal Python (hashlib + scipy.stats). Inputs are the tokenizer’s token IDs, the secret key, and the state count; outputs are the fingerprint ϕ , z -score, one-sided p -value, and a binary detection flag.